



BITS Pilani

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AIML for Robotics

Module 2

Components of Programmable Robotic System

Motion Model

Section Faculty:

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The course owner and the authors of this course deck, Prof. Raja vadhana and Dr.Sangeetha, are gratefully acknowledging all authors who made their course materials freely available online.

☐ Introduction

- ☐ Types of Robots
- ☐ Mobile Robots – Use case of Programmable solutions

☐ Components of Programmable Robotic System

- ☐ Perception
- ☐ Representation
- ☐ Motion

☐ Localization

- ☐ Taxonomy
- ☐ Types

☐ Mapping

- ☐ Feature Based
- ☐ Grid Based

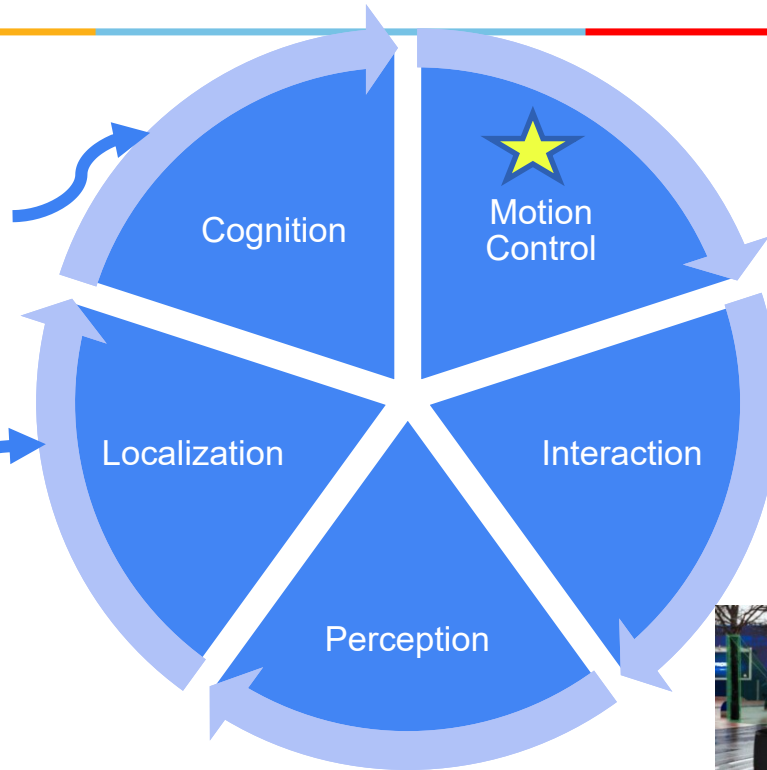
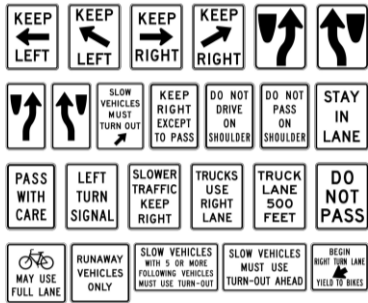
☐ Path Planning

- ☐ Graph Based
- ☐ Sample Based
- ☐ Evolutional Techniques

☐ Human – Robot Interaction

- ☐ Ethics
- ☐ Introduction to Visual Bot
- ☐ Introduction to Chat Bot

Perception & Control



Estimation of State

1. Percept Measurement Data - Momentary State of the Env

2. Motion Control Data - Change of State in Env

$$1. z_{t1:t2} = z_{t1}, z_{t1+1}, z_{t1+2}, \dots, z_{t2}$$

$$2. u_{t1:t2} = u_{t1}, u_{t1+1}, u_{t1+2}, \dots, u_{t2}$$

3. Prediction : Control Update : $p(x_t | z_{1:t-1}, u_{1:t})$:
Control $\rightarrow$ Infer $\rightarrow$ Perceive

Correction : $p(x_t | z_{1:t}, u_{1:t})$: Control $\rightarrow$ Perceive $\rightarrow$ Infer
(Measurement Update) : Predict $\rightarrow$ Infer

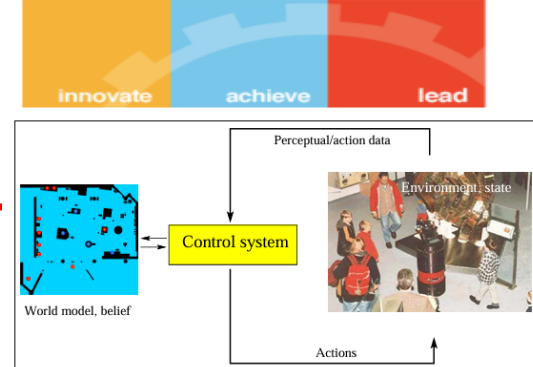


Figure 2.1 Robot Environment Interaction.

Even though odometers are sensors, we will treat odometry as control data, since its main information regards the change of the robot's pose.

State Estimation

Bayes Filter Algorithm

1. Initial Belief
2. **Control Update**
3. Measurement Update

Objective : Estimate the state of the door

State Space : {is_open, is_closed}

Action Space : { push, do_nothing}

Use case : Predict the state of the door, given robot's action & sensor data observed :

(?, do_nothing, sense_open, push, sense_open)
(?, A1, S2, A2, S3)

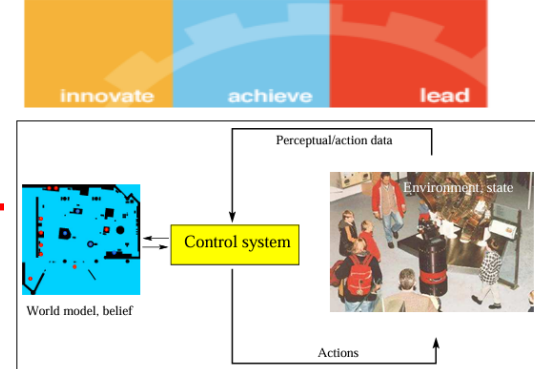


Figure 2.1 Robot Environment Interaction.

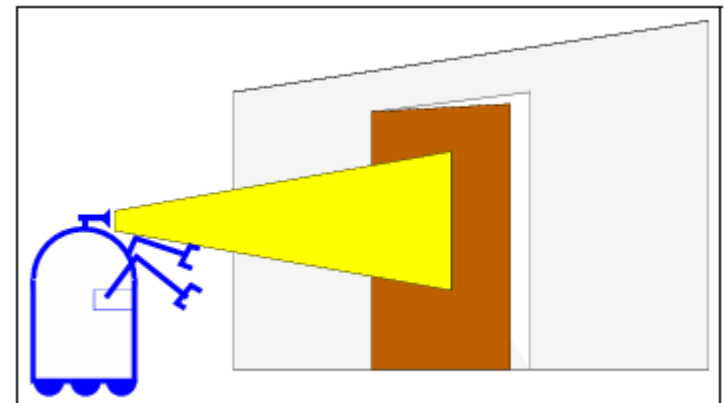


Figure 2.2 A mobile robot estimating the state of a door.

X_0	$P(X_0)$
is_open	0.5
is_closed	0.5

X_{t-1}	U_t	$P(X_t = \text{is_open})$	$P(X_t = \text{is_closed})$
is_open	push	1	0
is_closed	Push	0.8	0.2
is_open	do_nothing	1	0
is_closed	do_nothing	0	1

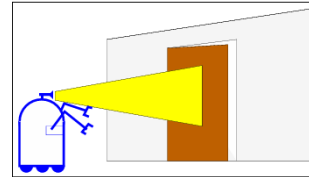
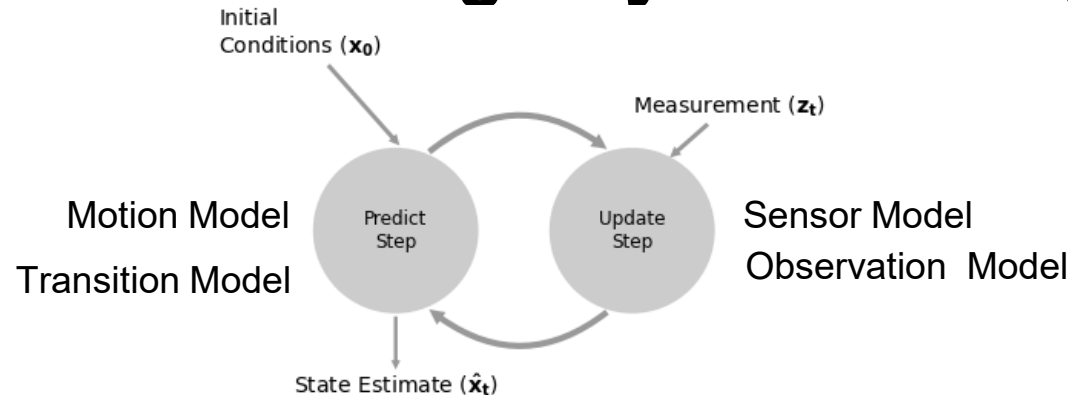


Figure 2.2 A mobile robot estimating the state of a door.

X_t	$P(Z_t = \text{sense_open})$	$P(Z_t = \text{sense_closed})$
is_open	0.6	0.4
is_closed	0.2	0.8

State Estimation using Bayes Filter Algorithm



Recap

1. Initial Belief
2. Control Update
3. Measurement Update

MOTION MODEL : $P(X_t | U_t, X_{t-1})$

X_{t-1}	U_t	$P(X_t = \text{is_open})$	$P(X_t = \text{is_closed})$
is_open	push	1	0
is_closed	Push	0.8	0.2
is_open	do_nothing	1	0
is_closed	do_nothing	0	1

X_0	$P(X_0)$
is_open	0.5
is_closed	0.5

X_t	$P(Z_t = \text{sense_open})$	$P(Z_t = \text{sense_closed})$
is_open	0.6	0.4
is_closed	0.2	0.8

$u_2 = \text{push}$ and $z_2 = \text{sense_open}$

$$\overline{\text{bel}}(X_2 = \text{is_open}) = 1 \cdot 0.75 + 0.8 \cdot 0.25 = 0.95$$

$$\overline{\text{bel}}(X_2 = \text{is_closed}) = 0 \cdot 0.75 + 0.2 \cdot 0.25 = 0.05$$

and

$$\text{bel}(X_2 = \text{is_open}) = \eta \cdot 0.6 \cdot 0.95 \approx 0.983$$

$$\text{bel}(X_2 = \text{is_closed}) = \eta \cdot 0.2 \cdot 0.05 \approx 0.017$$

Inference :

The robot believes that with 0.983 probability the door is open

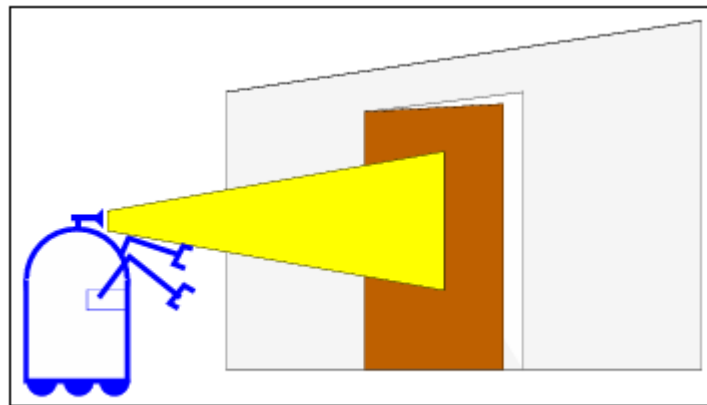
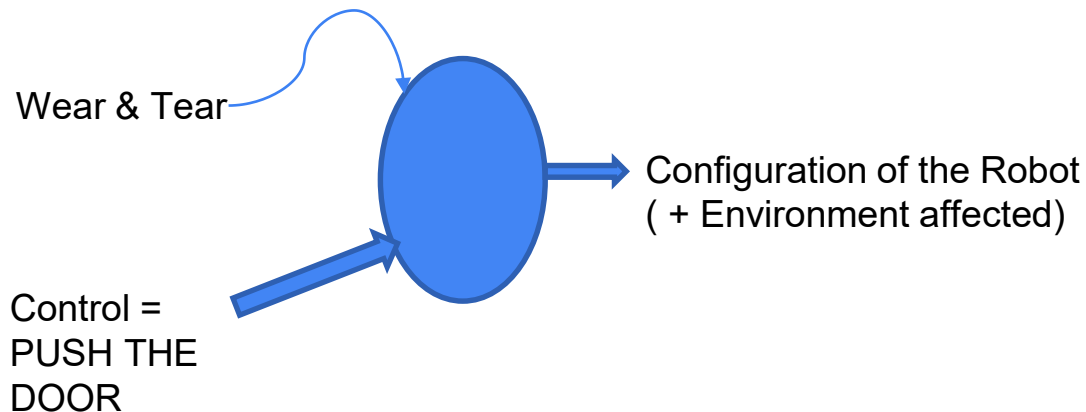


Figure 2.2 A mobile robot estimating the state of a door.

Link to class demo file : (Accessible only via the BITS email login . Students are advised download and work in the copy of the file)

<https://colab.research.google.com/drive/1voSWkZ2rE6y6vOI2mvHjqOtyldMgsm7m?usp=sharing>

Motion Model



1. Velocity

- Here the control is treated as the intended displacement from velocity commands (Translational, Angular)

2. Odometer

3. Gaussian

4. Sampling based

5. Vision

6. Language

Case :

Robot needs to plan before moving to avoid collision to obstacles. How to predict them before actually moving?

$$x_{t-1} = \begin{bmatrix} x = 5 \\ y = 0 \\ \theta = \pi/6 \end{bmatrix} u_t = \begin{bmatrix} V = 1m/s \\ w = 1m/s \end{bmatrix}$$

$$x_t = \begin{bmatrix} x' = 3 \\ y' = 1.5 \\ \theta' = 40^\circ \end{bmatrix}$$

$$P(X_t | U_t, X_{t-1})$$

1. Velocity
 2. **Odometer**
 3. Gaussian
 4. Sampling based
 5. Vision
 6. Language
- This model doesn't use control directly instead uses the wheel displacement (left, right). Conceptually the Delta change in (X, Y) can be inferred from the readings

Case :

Robot needs to estimate the distance to an obstacle while moving for precision navigation. How to predict them while actually moving?

$$x_{t-1} = \begin{bmatrix} x = 5 \\ y = 0 \\ \theta = \pi/6 \end{bmatrix} u_t = \begin{bmatrix} \overline{x_{t-1}} \\ \overline{x_t} \end{bmatrix}$$

$$x_t = \begin{bmatrix} x' = 3 \\ y' = 1.5 \\ \theta' = 40^\circ \end{bmatrix}$$

$$P(X_t | U_t, X_{t-1})$$

1. Velocity
2. Odometer
3. **Gaussian**
4. Sampling based
5. Vision
6. Language

➤ Here the mean displacement is form of control. As long as single sample from Gaussian is sufficient with unimodal noise to depict the result this works good.

More on this we will connect in upcoming modules wherever applicable

$$x_{t-1} = \begin{bmatrix} x = 5 \\ y = 0 \end{bmatrix} u_t = \begin{bmatrix} \mu x, stdx \\ \mu y, stdy \end{bmatrix}$$


$$x_t = \begin{bmatrix} x' = 3 \\ y' = 1.5 \end{bmatrix}$$


$$P(X_t | U_t, X_{t-1})$$

1. Velocity
 2. Odometer
 3. Gaussian
 4. **Sampling based**
 5. Vision
 6. Language
- Here the control is the intended displacement. If the noise is multimodal, this works by assuming multiple possible answers(hypothesis). Weighs each hypothesis based on the observation before filtering the best estimate. Refresh on our CS-2 discussion on the working of Particle filtering algorithm.

$$x_{t-1} = \begin{bmatrix} x = 5 \\ y = 0 \\ \theta = \pi/6 \end{bmatrix} u_t = \begin{bmatrix} \overline{x_{t-1}} \\ \overline{x_t} \end{bmatrix}$$

$$x_t = \begin{bmatrix} x' = 3 \\ y' = 1.5 \\ \theta' = 40^\circ \end{bmatrix}$$

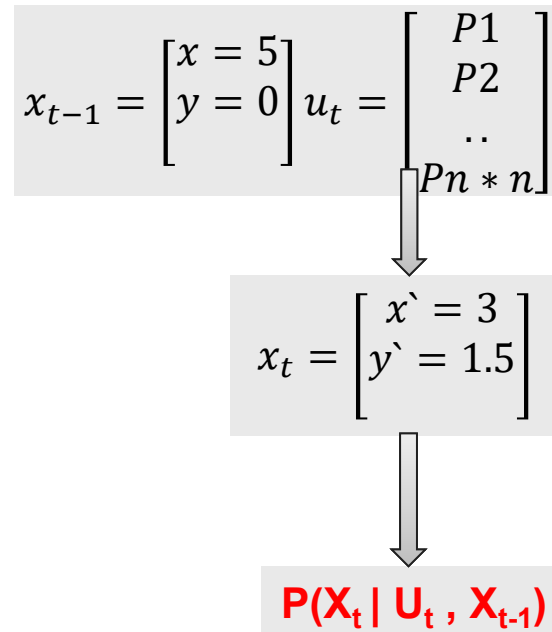
$$P(X_t | U_t, X_{t-1})$$

More on Particle Filters, we will revisit in upcoming modules under Module on Localization

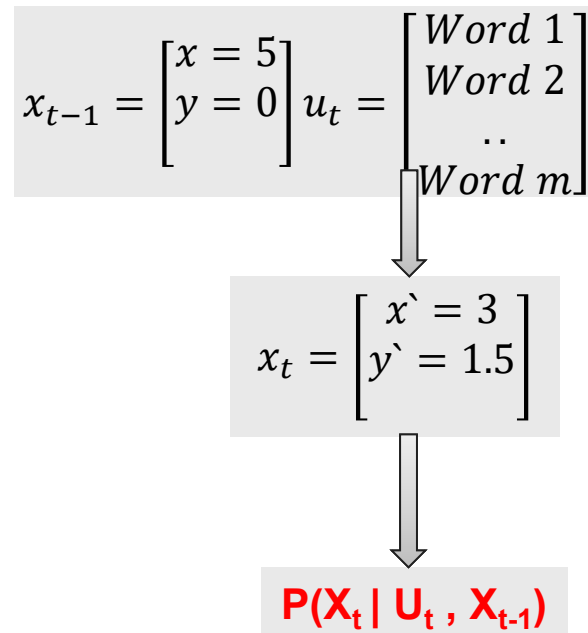
Motion Models



1. Velocity
 2. Odometer
 3. Gaussian
 4. Sampling based
 - 5. Vision**
 6. Language
- Here the motion commands is inferred from image inputs (pics or video).



1. Velocity
 2. Odometer
 3. Gaussian
 4. Sampling based
 5. Vision
 6. **Language**
- Here the motion commands is extracted from the pattern observed in the sequential language inputs / embedding (Text or Speech-to-Text converted).



Recap

1. Initial Belief
2. Control Update
3. Measurement Update

X_{t-1}	U_t	$P(X_t = \text{is_open})$	$P(X_t = \text{is_closed})$
is_open	push	1	0
is_closed	Push	0.8	0.2
is_open	do_nothing	1	0
is_closed	do_nothing	0	1

SENSOR MEASUREMENT MODEL : $P(Z_t | X_t)$

X_t	$P(Z_t = \text{sense_open})$	$P(Z_t = \text{sense_closed})$
is_open	0.6	0.4
is_closed	0.2	0.8

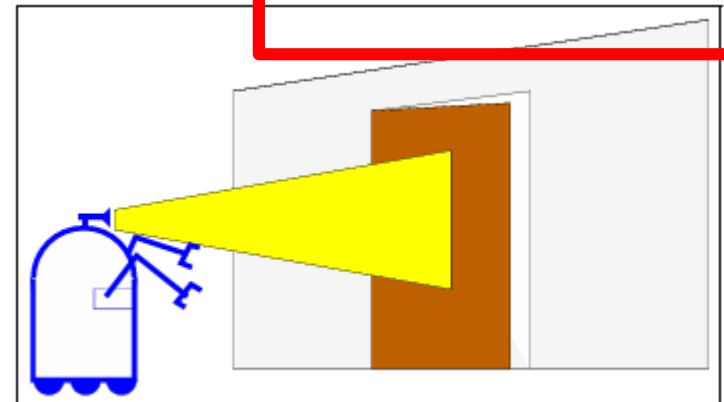


Figure 2.2 A mobile robot estimating the state of a door.

(Prof. Sangeetha will continue the Sensor Model)

Required Reading

- T1 Chapter #5, #6
- T2 Chapter #5, #6.1, #6.2

Next Session Plan

- ✓ Sensor Measurement Models
- ✓ Robot Localization